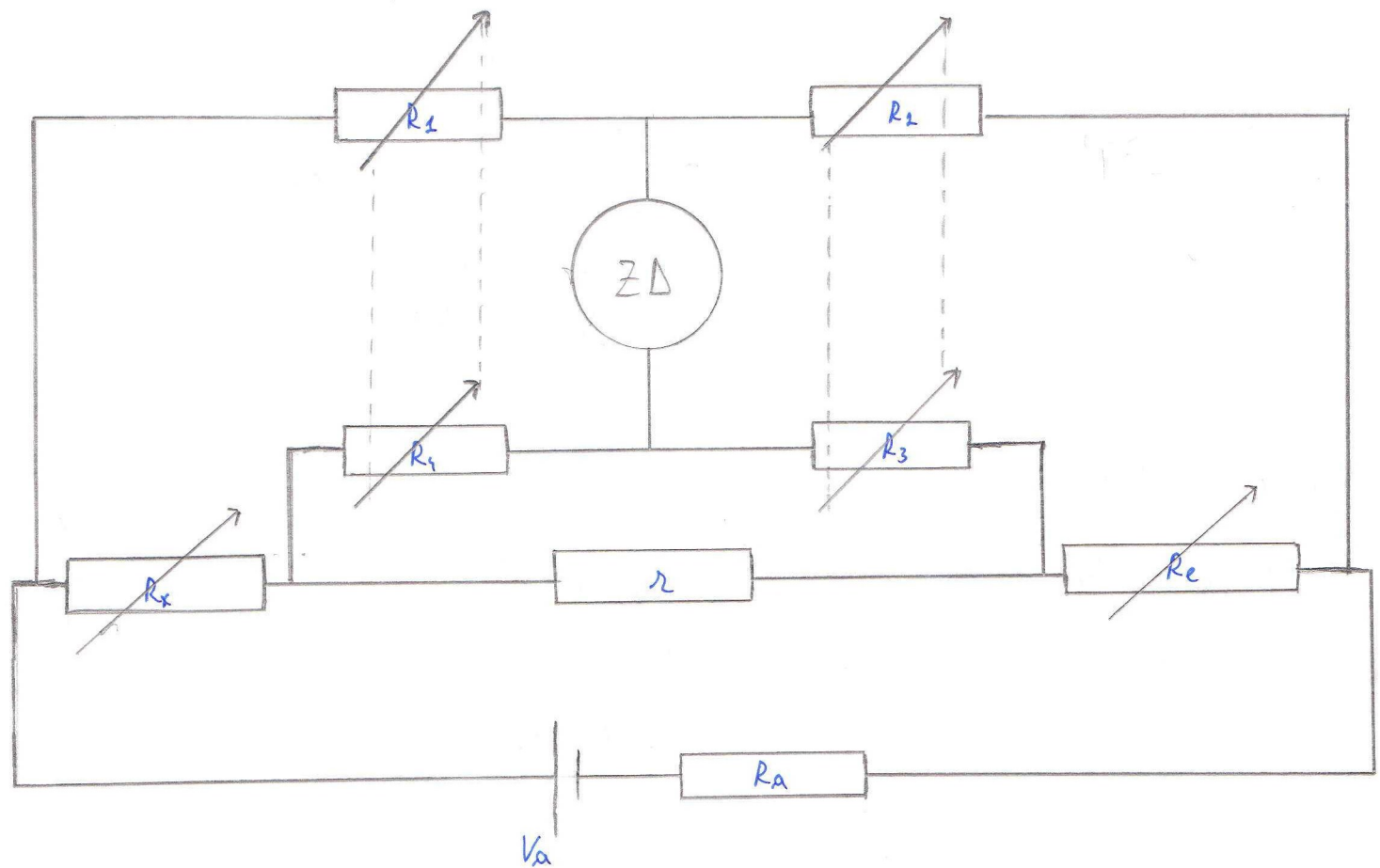
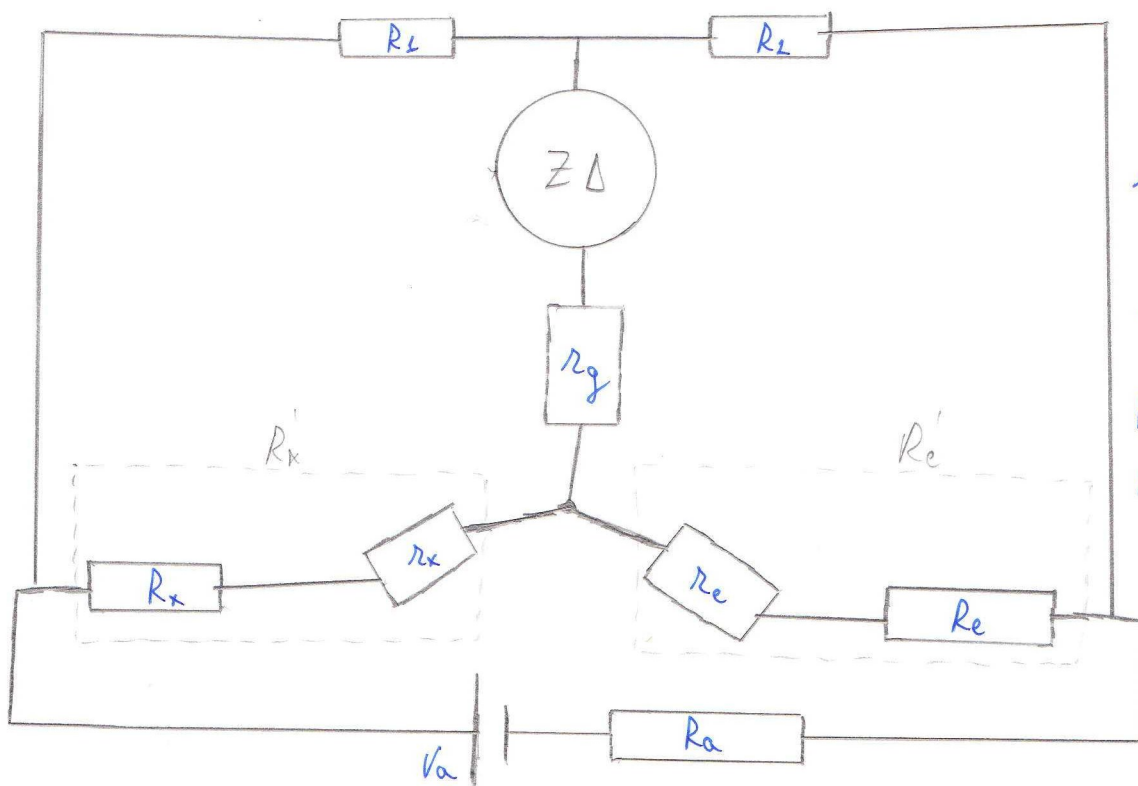


Sensitivity computation  
of  
Double D.C.  
(Thomson)  
Bridge

MUNTEAN  
MARIUS  
3021  
HOMEWORK 2



To calculate the relative sensitivity, the subcircuit  $\Delta r R_3 R_4$ , will be transformed in the equivalent  $\Delta$  star subcircuit.



$$R_x = \frac{r \cdot R_4}{r + R_3 + R_4}$$

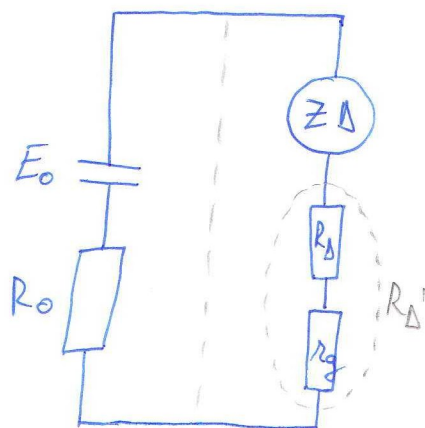
$$R_e = \frac{r \cdot R_3}{r + R_3 + R_4}$$

$$R_G = \frac{R_3 \cdot R_4}{r + R_3 + R_4}$$

$$R_x' = R_x + R_x$$

$$R_e' = R_e + R_e$$

Thevenin:



Relative sensibility:  $S_R = \lim_{\Delta R'_x \rightarrow 0} \frac{\Delta \alpha}{\Delta R'_x / R'_x} = R'_x \cdot \frac{d\alpha}{dR'_x}$

The relative sensibility is defined as a ratio between the output and the input of a circuit.

output: deflection of  $Z_\Delta$ ,  $\propto$

input: variation of  $R_x$

$$E_0 = V_a \cdot \left( \frac{R_2}{R_1 + R_2} + \frac{R_e'}{R_e' + R'_x} \right); \quad R_0 = \frac{R_1 \cdot R_2}{R_1 + R_2} + \frac{R_e' \cdot R'_x}{R_e' + R'_x}$$

$$I_\Delta = \frac{E_0}{R_0 + R'_0} = V_a \cdot \frac{R_2 \cdot R'_x - R_1 \cdot R_e'}{a \cdot R'_x + b} \quad \left( = \frac{R_1 \cdot R_2 (R_e' + R'_x) + R_e' \cdot R'_x (R_1 + R_2)}{R'_x (R_1 + R_2) + R_1 \cdot R_e' + R_2 \cdot R_e'} \right)$$

$$\alpha = S_I \cdot I_\Delta = S_I \cdot V_a \cdot \frac{R_2 \cdot R'_x - R_1 \cdot R_e'}{a \cdot R'_x + b}$$

$$S_R = R'_x \cdot \frac{d\alpha}{dR'_x} = R'_x \cdot S_I \cdot V_a \cdot \frac{R_2(a \cdot R'_x + b) - (R_2 \cdot R'_x - R_1 \cdot R_e') \cdot a}{(a \cdot R'_x + b)^2}$$

The sensibility is highest if  $\begin{cases} R_1 = R_2 = R'_x = R_e' \\ R_2 \cdot R'_x = R_1 \cdot R_e' \end{cases} \Rightarrow$

$$\Rightarrow S_R \approx \frac{R_2 \cdot R'_x}{(a \cdot R'_x + b)} \cdot S_I \cdot V_a \cdot \frac{R_2 \cdot R'_x}{(R_1 + R_2) \cdot R'_x + R_1 \cdot R_e' + R_2 \cdot R_e'}$$

even more detailed: 
$$S_R \approx \frac{R_2 \cdot \left( R_x + \frac{r \cdot R_4}{r + R_3 + R_4} \right)}{(R_1 + R_2) \left( R_x + \frac{r \cdot R_4}{r + R_3 + R_4} \right) \left( R_e + \frac{r \cdot R_3}{r + R_3 + R_4} \right)} \cdot S_I \cdot V_a$$